DiscFlow Pump | spec sheet

Tesla Lab Product Design Engine - numbers computed in Python, not guessed

A bladeless pump that doesn't clog and is gentle on whatever it moves.

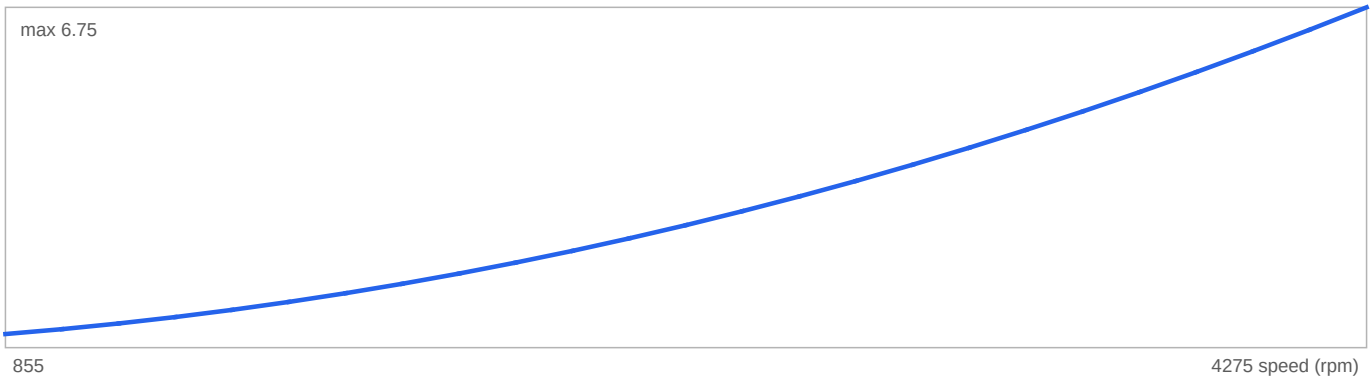
Derived from the claim that WORKS: Smooth spinning discs can exchange energy with a fluid by boundary-layer drag, without blades.



Key numbers

- 60 L/min at 3 m head from a 51 mm stack of 27 discs at 2850 rpm
- Gaps 0.8 mm pass solids up to that size, no blades to clog or bruise fish
- 117 W shaft power, 180 W motor (at an assumed 25% efficiency)
- Costs about $77/yr more electricity than a 50% centrifugal pump (estimate)
- Prototype parts ~ $264 (estimate)

head (m) vs speed (rpm)



Requirements

Flow	**60 L/min**	req
Head	**3 m**	req
Motor speed	**2850 rpm**	req

2-pole induction motor on 50 Hz mains (Malaysia/Singapore)

Rotor

Tip speed	**7.67 m/s**	calc
head coefficient psi = 0.5 (assumed)		
Disc diameter	**51.4 mm**	calc
Inlet (eye) diameter	**20.6 mm**	calc
Boundary-layer thickness sqrt(nu/w)	**53 um**	calc
Disc gap	**0.8 mm**	calc
set by solids - wider than the efficiency optimum, the price of not clogging		
Number of discs	**27**	calc
26 flow gaps		
Stack length	**34.3 mm**	calc
Gap Reynolds number wb^2/nu	**229**	calc

Power

Hydraulic power rhogQH	**29.3 W**	calc
Hydraulic efficiency	**25 %**	assumed
published range 20-30% for disc pumps		
Shaft power	**117 W**	calc
Motor (next standard size, 15% margin)	**180 W**	calc

Running cost

Energy per day	**3.52 kWh**	calc
Electricity per year	**154 USD**	est.
at $0.12/kWh (assumed)		
Same duty, centrifugal pump at 50%	**77 USD/yr**	est.

Bill of materials (estimates)

180 W 2-pole motor (IP55)	**1 x $85.00**	$85
Stainless 304 discs, laser-cut, 0.5 mm	**27 x $1.50**	$40
Spacers / washers set	**1 x $8.00**	$8
Shaft, bearings, mechanical seal	**1 x $35.00**	$35
Clear polycarbonate volute housing (machined)	**1 x $60.00**	$60
Hose fittings, clamps, mounting bracket	**1 x $15.00**	$15
Motor starter / switch, cable	**1 x $20.00**	$20

Total (estimate): $264

Rough estimates from typical 2025-26 distributor / retail list prices for one-off prototype quantities, not quotes. Verify before you buy.

Honest limits

Hydraulic efficiency is around 20-30% (S4), well below a good centrifugal pump. It's worth it only where clogging, shear or noise matter more than electricity cost.

Caveats

- First-order sizing: the head coefficient and efficiency are assumptions to calibrate on a prototype.
- Laminar-flow assumption between discs; check the gap Reynolds number.
- Worth it only where clogging, shear or noise matter more than electricity cost.

Render prompt (generated from these numbers)

Photoreal product render, 3/4 view, soft daylight, shallow depth of field, no text. A compact bladeless disc pump: a clear polycarbonate housing about 82 mm across shows a stack of 27 thin polished stainless-steel discs, each 51 mm in diameter with 0.8 mm gaps, total stack about 34 mm long, driven by a small 180 W grey electric motor on the same shaft. A central inlet hose and a tangential outlet hose; mounted beside a small aquaculture pond with water droplets on the housing.

References

- Tesla, US 1,061,206 Turbine (1913) https://patents.google.com/patent/US1061206A/en

- Infinity Turbine, Tesla disc pump review (20-30% hydraulic efficiency)
https://www.infinityturbine.com/tesla-pump-review-article-by-infinity-turbine.html
- Heng et al., Tesla bladed pump preliminary study, Energies 13(18):4873 (2020) https://www.mdpi.com/1996-1073/13/18/4873